Statistics Tiers Reference

No	Level	Topics	Key Concept Examples	Example of Use in Research
1	**Foundations**	• Descriptive statistics • Probability distributions • Sampling & standard error • Data types (nominal, ordinal, continuous) • Central tendency & spread (mean, median, SD, IQR)	• **Mean / Median / SD:** summarise a set of benchmark run times across 30 trials • **Normal vs skewed distribution:** runtime data is often right-skewed due to occasional slow outliers • **Standard error:** how much your sample mean might vary if you repeated the same experiment • **IQR:** robust spread measure when outliers are present — preferred over SD for skewed data	• **Reporting results:** 'Algorithm A achieved a median accuracy of 92% (IQR: 89–94%) across 30 runs' • **Choosing the right test:** understanding your distribution determines whether parametric or non-parametric tests are appropriate • **Exploratory analysis:** always visualise and summarise your data before applying any test
2	**Inferential Statistics**	• Hypothesis testing (null vs alternative) • p-values & statistical significance • Confidence intervals • Effect size (Cohen's d, η^2) • t-tests (independent & paired)	• **p-value:** probability of results this extreme if there were truly no difference — NOT the probability your hypothesis is true (common misconception) • **Effect size (Cohen's d):** d = 0.2 small, 0.5 medium, 0.8 large — tells you if a difference matters, not just whether it exists • **95% CI:** range within which the true parameter likely falls; a CI that excludes zero suggests a real effect • **Paired t-test:** comparing the same system before and after an optimisation on identical workloads	• **Comparing two models:** 'Is Model B's F1 score significantly higher than Model A's, or within noise?' → independent-samples t-test • **Optimisation evaluation:** paired t-test on response times before/after a system change, using the same benchmark suite • **Key principle:** always report effect size alongside p-value — statistical significance ≠ practical importance
3	**Experimental Design**	• Control & treatment conditions • Randomisation & blinding • Repeated measures & trial runs • Identifying & controlling confounds • Baselines & ablation studies	• **Confounding variable:** testing Algorithm A on a faster machine than B makes hardware a confounder — results are uninterpretable • **Repeated measures:** running 30 trials per condition and averaging reduces random noise in results • **Ablation study:** remove one component at a time (e.g. attention module) to isolate each contribution • **Baseline:** compare against a simple/existing method to demonstrate your contribution adds real value	• **ML benchmarking:** fix dataset, hardware, random seeds — vary only the model; this is the minimum standard for a credible comparison • **Neural network paper:** ablation study trains model with/without each proposed module to show each part's isolated effect • **Reproducibility:** venues increasingly require full experimental conditions (seeds, hardware, splits) to be reported
4	**Applied to CS**	• ANOVA (one-way, two-way) • Non-parametric tests (Mann-Whitney U, Wilcoxon, Kruskal-Wallis) • Regression (linear, logistic, multiple) • Multiple comparisons correction (Bonferroni, Holm) • Bootstrap & permutation tests	• **Mann-Whitney U:** non-parametric alternative to t-test; preferred when runtimes or accuracy scores are not normally distributed • **Kruskal-Wallis:** compare 3+ algorithm variants without assuming normality — the non-parametric equivalent of one-way ANOVA • **Bonferroni correction:** testing 5 comparisons at $\alpha = 0.05$ → adjust threshold to 0.01 per test to control false positive rate • **Logistic regression:** predict binary outcome (e.g. bug present/absent) from software metrics as predictors	• **Comparing 4 sorting algorithms:** Kruskal-Wallis rather than multiple t-tests — avoids inflated Type I error from repeated testing • **Software defect prediction:** logistic regression with code metrics (LOC, cyclomatic complexity) as predictors of defect probability • **Runtime benchmarks:** Mann-Whitney U preferred over t-test because execution times are rarely normally distributed
5	**Human Subjects (if applicable)**	• Survey design & Likert scales • Inter-rater reliability (Cohen's κ, Krippendorff's α) • Qualitative–quant mixed methods • Thematic analysis basics • Ethical considerations in data collection	• **Likert scale:** 1–5 or 1–7 ordinal rating; treat as ordinal (report median, not mean) unless the scale has been formally validated • **Cohen's κ:** agreement between two raters beyond chance; $\kappa > 0.6$ = substantial, $\kappa > 0.8$ = near-perfect • **Mixed methods:** combining interview themes with quantitative survey scores to produce richer, triangulated findings	• **Usability study on a developer tool:** participants rate ease of use on a 5-point scale; Mann-Whitney U compares two interface designs • **Code review study:** two researchers independently label comments as 'helpful/unhelpful'; Cohen's κ checks agreement before using labels as ground truth • **Scope note:** only needed if your research involves user studies, interviews, or human-labelled data